



Tool for Bolts Installation in the D11R D11N D11T Sprocket Segments

**Maintenance and
Repair**

Application

**Component Life
Management**

**Component
Renewal (CRC)**

**MARC
Management**

Tool for Bolts Installation in the D11R D11N D11T Sprocket Segments.....	0
1.0 Introduction	1
2.0 Best Practice Description.....	2
3.0 Implementation Steps	3
4.0 Benefits.....	3
5.0 Resources Required	3
6.0 Supporting Attachments / References	4
7.0 Related Best Practices	4
8.0 Acknowledgements.....	5

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices.” Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

1.0 Introduction

The Installation of the sprocket segments on TTT is a task in which we must be especially careful when tightening the bolts since such failure could cause an unscheduled shutdown of the machine and risks the safety of the technical personnel performing the task. Technicians are exposed to pinch points and arm fatigue.

Currently for the torque procedure of the bolts a lever resting on the final drive is used in order to hold the nuts exposing the technician to a possible accident and it ensures the correct torque on the bolts is used. Fig.1.

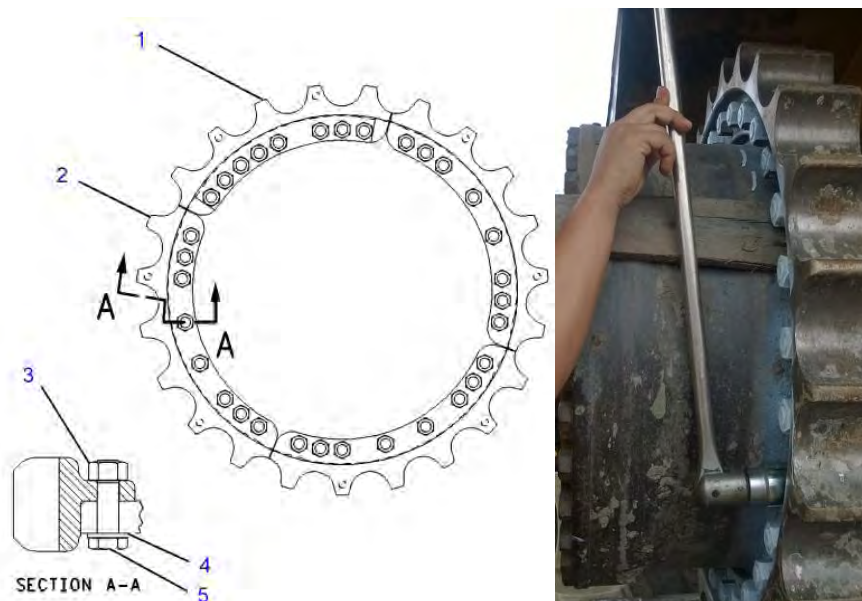


Fig. 1. Former old bolt torque process.

The figure above shows the way the technicians are exposed when seizing the wrench holding the nuts.

This Best Practice provides the design and manufacture of a tool for fastening the nuts when performing the bolts torque procedure.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Tool for Bolts Installation in the D11R D11N D11T Sprocket Segments	DATE 9/11/2015	CHG NO 00	NUMBER 1504-1.01-1371

2.0 Best Practice Description

This applies when changing the sprocket segments in D11N, D11R and D11T Tractors where there have been delays in the proper installation of the bolts because there is no tool to hold the nut while tightening the bolts.

Geometrically, the tool has a "U" shape, in one of its end, it has a 1 1/4 " and 4 1/4" diameter bar in length and in the other end it has a diameter of 2 1/4 " and a length of 1 3/8" bored concentrically with the silhouette of the head of the bolt to be installed. The two ends are joined by a 1" plate triangle shaped having rounded edges due to the difference in diameters of the two ends. Made entirely of ASTM A36 steel, this tool also has a handle so that the technician can comfortably hold it with his hand when installing or transporting.

In Fig. 3. We can see the way it holds the head of the bolt and it takes advantage of the segments teeth in order to lock itself and thus it reduces the exposure of the technicians to possible accidents and it improves the torque of the bolts reducing the installation time by up to 40 %.

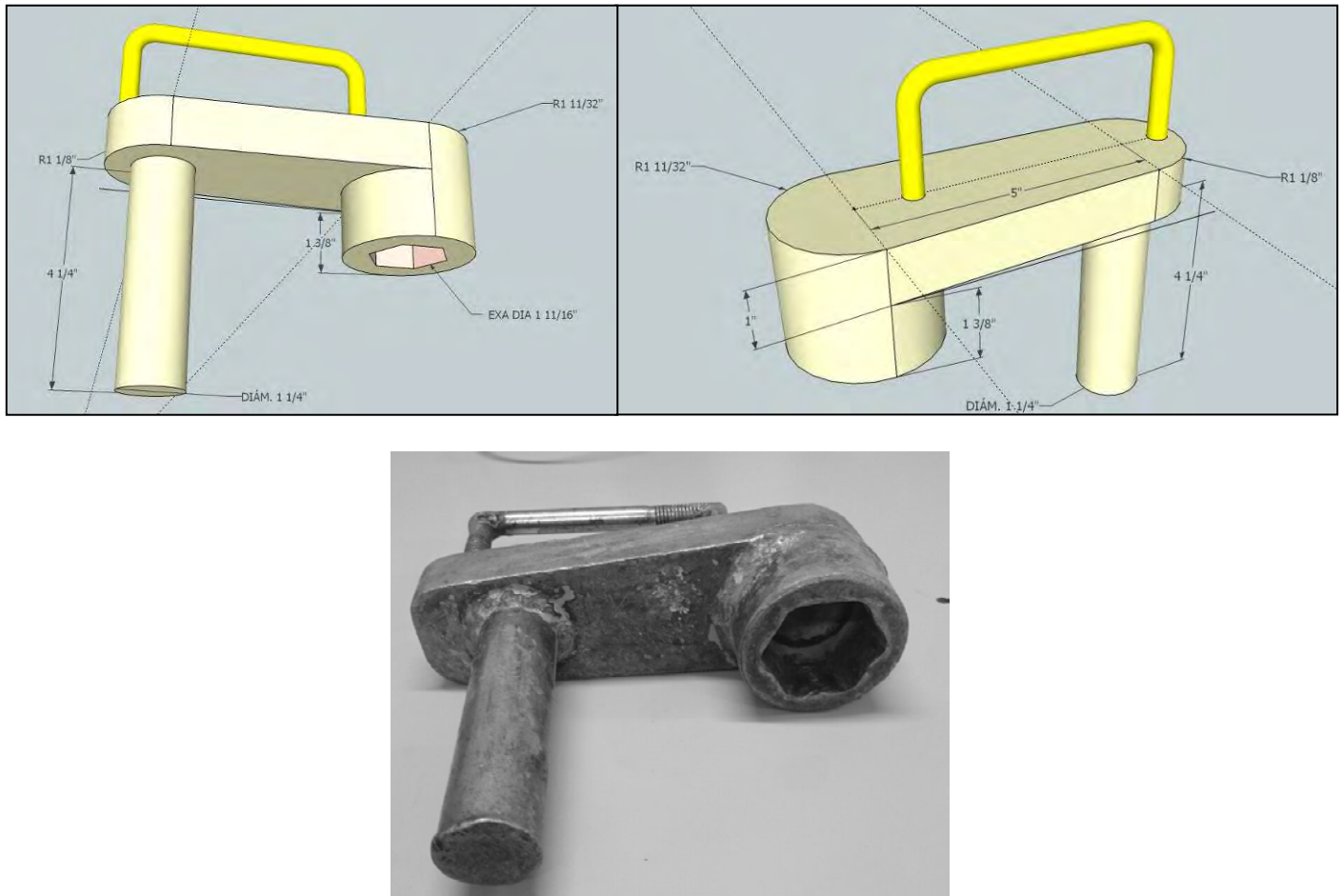


Figure 2. Sprocket segments bolts holding tool.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Tool for Bolts Installation in the D11R D11N D11T Sprocket Segments	DATE 9/11/2015	CHG NO 00	NUMBER 1504-1.01-1371

3.0 Implementation Steps

- o Design of the tool.
 - EDITOR'S NOTE: Caterpillar recommends considering making this tool with a 1" square drive so it can work on smaller tractors.
- o Manufacturing of the tool.
- o Operation of the tool (see Figure 3)
 - o From the inside of the sprocket, place the largest diameter end of the tool in order to hold the bolt head with the bored silhouette.
 - o Make sure the opposite end of the tool (1 ¼ " bar) is placed in one of the spaces between the sprocket teeth.
 - o Install the torque meter along with the adapters required to tighten the nut on the outside of the sprocket, make sure when rotating the tool that it is locked with one of the sprocket teeth thereby preventing the rotation of the bolt.



Figure 3. Operation of the tool for applying torque the screws segments sprocket

4.0 Benefits

- It improves the safety of the task.
- It reduces the installation time of the sprocket segments.
- It improves the torque application.
- There is higher availability of the equipment.

5.0 Resources Required

- Design drawings (see Figure 2)

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Tool for Bolts Installation in the D11R D11N D11T Sprocket Segments	DATE 9/11/2015	CHG NO 00	NUMBER 1504-1.01-1371

6.0 Supporting Attachments / References

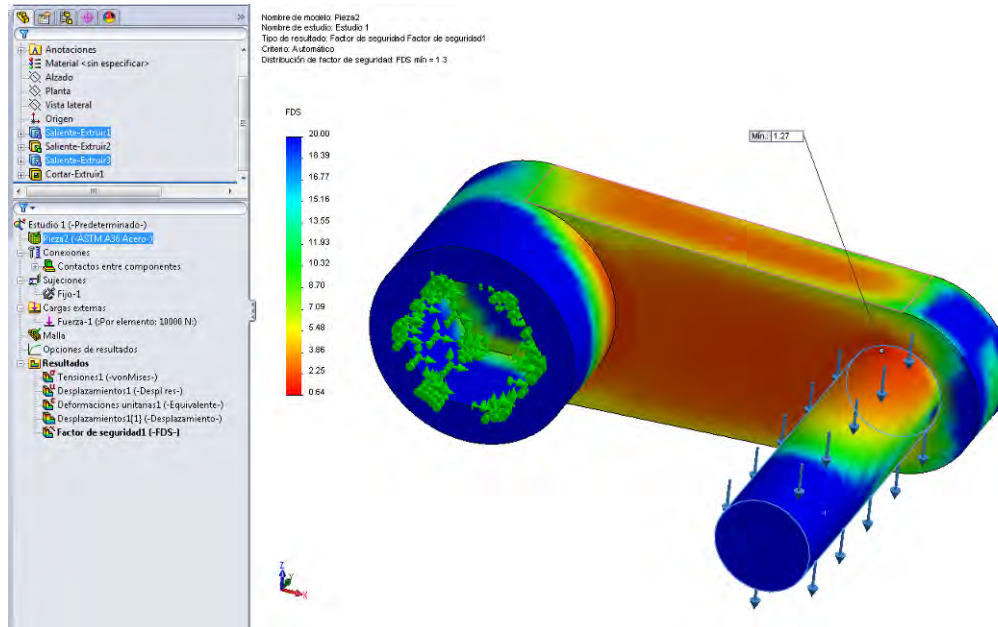


Figure 4. Analysis of resistance; load = 1 ton; FS = 1.27

7.0 Related Best Practices

None

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Tool for Bolts Installation in the D11R D11N D11T
Sprocket Segments

DATE
9/11/2015

CHG
NO
00

NUMBER
1504-1.01-1371

8.0 Acknowledgements

Jorge Monterroza
 Drummond Ltd Projects Undercarriage Shop Foreman.
 RELIZAN_CAT Mining Division
 Email: Jorge.monterroza@relianz.com.co

Martin Gonzales
 DRUMMOND Ltd Tractors fleet Planner.
 Email: mgonzalez@drummonddltd.com

Marcos Mendoza
 DRUMMOND Ltd Tractors Technical.
 Email: mmendozamar@gmail.com

Juan Valets
 DIMANTEC Project Line Supervisor

Samuel Vellojin
 RELIANZ Quality Engineer .
 Email: samuel.vellojin@relianz.com.co

Checking by:

Luis Alberto Fernandez Chica | BPs Coordinator and Development Engineer
 RELIANZ Mining Solutions | Tel: (5) 3361200, Ext: 5347 | Cellphone: 3013724303 |
luisalberto.fernandez@relianz.com.co

Edgard E. Romero Avena | Translator
edgard.romero@relianz.com.co

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Tool for Bolts Installation in the D11R D11N D11T Sprocket Segments	DATE 9/11/2015	CHG NO 00	NUMBER 1504-1.01-1371
--	-------------------	-----------------	--------------------------